

Daniel Lofeodo

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Experience

2025-04 – Machine Learning Engineer (Fixed-Term Contract)

2026-05 *Hydro-Québec, Montreal, Canada*

Stress-tested 3 hydro-dams in simulation against 1-in-100 to 1-in-10,000-year floods for the internal safety team, using RL/SL imitation learning agents that mimic operator behaviour.

- Built a **safety-bounded PPO agent** that stayed in bounds across all 9 extreme flood scenarios, cutting spilled water volume **83%** vs. the legacy HEC-ResSim baseline.
- Wrote the dam simulator and fixed stalled training by replacing a discontinuous reward with a continuous one that removed gradient reversal at the bounds.
- Benchmarked XGBoost, PatchTST, TimeXer and Chronos; the best supervised model breached bounds in **7 of 9** extreme scenarios despite strong historical accuracy.
- Built the Databricks/Spark ETL over SCADA, weather and dam data at mixed resolutions, cutting runtime from **45 to 10 minutes**.

2021-06 – Software Engineer

2024-08 *Creaform, Montreal, Canada*

Ultra precision 3D scanners and inspection software; team of 8 within about 30 engineers, multi-million line C++/C# codebase with nightly builds and strict peer review.

- Co-built the corrosion-assessment module of **VXintegrity** (**near seven-figure annual revenue**), color-mapping per-triangle mesh deviation from auto-fitted GD&T reference geometry.
- Built a **TBB-parallelized** scan-to-CAD deviation heat map with customizable color scales, and wrote low level geometry algorithms for form tolerances.
- **Solely owned** the VXelements UI/UX revamp **after one year on the job**, including customizable layouts and profiling that cut a GD&T freeze by at least half.
- Owned the C++/CLI WinForms to C#/WPF migration of two products, and built solo a WPF overlay for on-cursor sketch input over a legacy WinForms viewer.

Education

2024-08 – M.Sc. Computer Science (Machine Learning)

2026-05 *Montreal Institute for Learning Algorithms, University of Montreal, Montreal, Canada*

- 4.1/4.3 GPA
- Professional master's; internship report on the Hydro-Québec RL work

2017-08 – B.Eng. Mechanical Engineering

2021-05 *Concordia University, Montreal, Canada*

- Dean's List (2021)

Spotlight Projects

2026-05 – Latent SpaceMail – Autonomous Multi-Agent Newsletter Pipeline

Present Live weekly AI briefing: **48 subscribers, 70%+ open rate** (SendGrid).

- Built a **6-agent pipeline** plus health check on GCP (Cloud Run, Pub/Sub, Firestore) with race-safe fan-in; Claude Haiku screens ~800 articles and up to 500 papers weekly.
- Added failure recording, stale-run alerts, and retry/backoff around arXiv throttling, and shipped a bilingual Firebase-auth subscriber site.

2026-04 – Perspective Blind – 3D Spatial Reasoning Benchmark for VLMs

2026-05 VLM benchmark on imagined viewpoints (5 models, ScanNet).

- Designed a benchmark with paired views sharing one analytic ground truth, separating accuracy from cross-view consistency.
- Found GPT-4o's lateral accuracy falls from **83.6% to 8.1%**, below chance, as viewpoint misalignment grows; it is only 11.6% consistent across views.

Technical Skills

Programming Languages:

Python, C++, C#, SQL, Java, JavaScript, TypeScript

AI / ML:

PyTorch, AutoML, scikit-learn, xarray, XGBoost, Pandas, Numpy, stable-baselines3, gymnasium, generative AI, machine learning, neural networks, LLMs, prompt engineering, retrieval-augmented systems, reinforcement learning (PPO, SAC), Chronos, forecasting

Data science / Mathematics:

Statistical analysis, EDA, ETL, algorithms, linear algebra, Apache Spark

Visualization Frameworks:

matplotlib, Seaborn, Plotly

Web/App Frameworks:

Streamlit, Flask, FastAPI, React

MLOps:

Azure, GCP, AWS, GitHub, Databricks, Docker, Weights & Biases, Tensorboard, Cloud Pub/Sub, Cloud Firestore, Cloud Run, CI/CD

Environments:

Windows, WSL, Linux

Project management:

Asana, Jira, Teams

Certifications

- AWS MLA (Cloud-native ML)
- Claude Code in Action
- Complete Modern C++ (C++11/14/17)

Conferences

- **Autonomous Hydroelectric Dam Operation under Extreme Flooding**, Mila Annual Conference

Languages

- English (Native)
- French (Native)
- Italian (Advanced - C1)